PHY101- GENERAL PHYSICS I
1st SEM, 2021/2022
WEEK 6 LECTURE 1

CHAPTER FOUR

KINEMATICS

Average speed, s

$$= \frac{distance}{time} = \frac{d}{t} \qquad 4.1$$

Average velocity, v

$$\frac{displacement}{time} = \frac{\Delta \mathbf{x}}{\Delta t} = \frac{x - x_0}{t - t_0} = \bar{v} \qquad 4.2$$

Instantaneous velocity, v

$$= \frac{d\mathbf{x}}{dt} = \lim_{t \to 0} \frac{x - x_0}{t - t_0} \qquad 4.3$$

Average acceleration, a

$$a = \frac{velocity}{time} = \frac{\Delta v}{\Delta t} = \frac{v - v_0}{t - t_0} = \bar{a} \qquad 4.4$$

$$= \frac{d}{dt}(\vec{v}) = \frac{d}{dt}\left(\frac{d\vec{x}}{dt}\right) = \frac{d^2\vec{x}}{dt^2}$$

Accelerated Motion:

Acceleration can occur when there is:

(a) a change in magnitude of the velocity but not the direction

(b) a change in the direction of the velocity but not the magnitude

(c) a change in both the magnitude and direction

Example 4.1 Mr. Tetteh and his driver are traveling to Cape Coast in a Toyota pickup which is moving at 90 km/h down a straight highway. The driver of the pickup sees an accident in the distance and slows down to 40 km/h in 5.0 s. What is the average acceleration of the Toyota pickup?

Solution: Given the initial and final velocities and the time duration

$$u = v_i = v_0 = (90 km/h) = (25 m/s)$$

$$v = v_f = (40 km/h) = (11 m/s)$$

$$t = 5.0 s$$

We are solving to find: average acceleration,

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v - v_0}{\Delta t} = \frac{v - u}{\Delta t} = \frac{v_f - v_i}{\Delta t}$$

$$= \frac{11m/s - 25m/s}{5.0s} = -2.8m/s$$

NB: -ve acceleration => deceleration

velocity in components

$$\bar{v} = \frac{\text{displacement}}{\text{time}} = \frac{\Delta \mathbf{r}}{\Delta t} \qquad 4.5$$

$$\bar{v} = \frac{\Delta x\mathbf{i} + \Delta x\mathbf{j} + \Delta x\mathbf{k}}{\Delta t} = \frac{\Delta x}{\Delta t}\mathbf{i} + \frac{\Delta y}{\Delta t}\mathbf{j}\frac{\Delta z}{\Delta t}\mathbf{k} \qquad 4.6$$

$$\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \Rightarrow \Delta\vec{r} = \Delta x\mathbf{i} + \Delta y\mathbf{j} + \Delta z\mathbf{k}$$

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} \qquad 4.7$$

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k} \qquad 4.8$$

$$\mathbf{v} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k} \qquad 4.9$$

$$v_x = \frac{dx}{dt}, v_y = \frac{dy}{dt}, v_z = \frac{dz}{dt}$$

Acceleration in components

$$\bar{a} = \frac{\text{change in velocity}}{\text{time interval}} = \frac{v_2 - v_1}{\Delta t} = \frac{\Delta v}{\Delta t}$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}) = \frac{dv_x}{dt}\mathbf{i} + \frac{dv_y}{dt}\mathbf{j} + \frac{dv_z}{dt}\mathbf{k}$$

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}, a_y = \frac{dv_y}{dt} = \frac{d^2y}{dt^2}\ a_z = \frac{dv_z}{dt} = \frac{d^2z}{dt^2}$$

Kinematic equations

$$\mathbf{v} = \frac{\mathbf{s}}{t} \qquad \mathbf{s} = \frac{1}{2}(\mathbf{v}+\mathbf{u})t \qquad \mathbf{a} = \frac{\Delta \mathbf{v}}{\Delta t} = \frac{\mathbf{v}-\mathbf{u}}{\Delta t}$$

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\mathbf{s} = \mathbf{x} = \mathbf{u}t + \tfrac{1}{2}\mathbf{a}t^2$$

$$v^2 = u^2 + 2as$$

Derivation of the equations of motion

$$\mathbf{a} = \frac{\mathbf{v} - \mathbf{u}}{t}$$

$$v = u + at$$

$$\mathbf{s} = \frac{1}{2}(\mathbf{v} + \mathbf{u})t$$

$$s = \tfrac{1}{2}(u + u + at)t = ut + \tfrac{1}{2}at^2$$

$$s = \left(\frac{v + u}{2}\right)\left(\frac{v - u}{a}\right)$$

$$t = (v - u)/a$$

$$s = \left(\frac{v+u}{2}\right)\left(\frac{v-u}{a}\right) \rightarrow s = \left(\frac{v^2-u^2}{2a}\right)$$

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as$$

END